

NAME:.....

CLASS: .....

## SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

### HSC ASSESSMENT TASK 2

MARCH 2007

### MATHEMATICS

Time Allowed : 70 minutes

**Instructions:**

- Write your name and class at the top of each page
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new page.

| Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Total |
|------------|------------|------------|------------|------------|-------|
| /12        | /12        | /12        | /12        | /10        | /58   |

**QUESTION 1****Marks****a)** Differentiate the following:

i)  $y = x^3 + 4x^2 + 2$  **1**

ii)  $y = \frac{3x}{x+2}$  **2**

iii)  $y = (2x+1)^4$  **2**

**b)** Find the gradient of the tangent to the curve  $y = 4x^3 + x$  at the point ( 1, 5) **2**

**c)** Find:

i)  $\int x^4 + 3x^2 \, dx$  **1**

ii)  $\int (x-5)(x+4) \, dx$  **2**

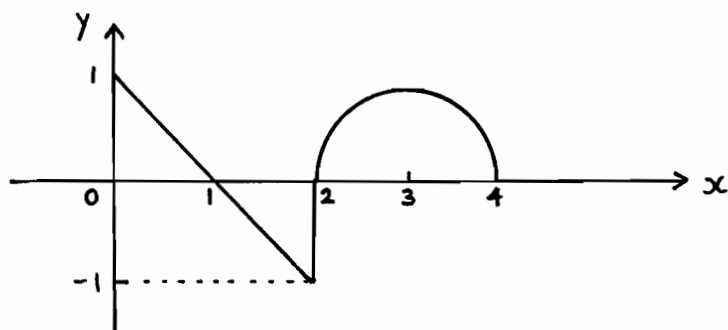
iii)  $\int \frac{x^3 - 3x^4}{x^2} \, dx$  **2**

## QUESTION 2

Marks

- a) Find the exact value of  $\int_0^4 f(x) dx$  given

2

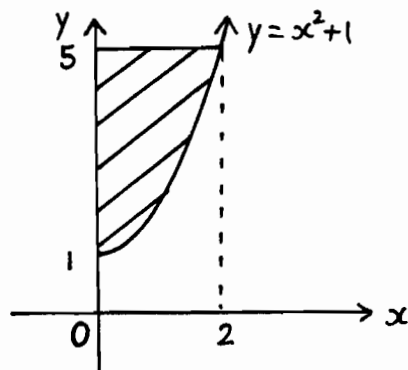


- b) Find  $\int (2x-1)^5 dx$

2

- c) The sketch shows an arc of the curve  $y = x^2 + 1$ .

3



Calculate the shaded area.

- d) The gradient of a curve is given by

$$\frac{dy}{dx} = 3x^2 - 12$$

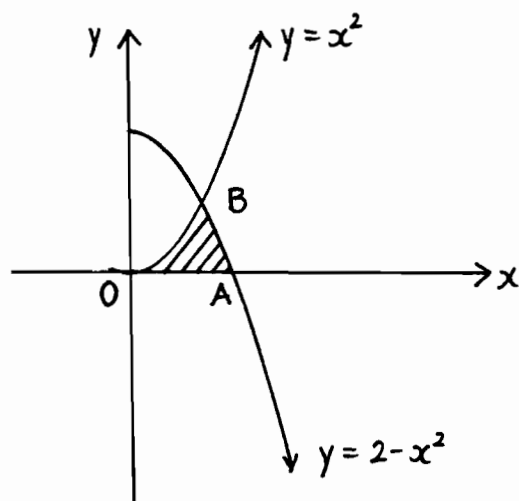
- i) Find  $\frac{d^2y}{dx^2}$  1
- ii) Find the values of  $x$  for which the curve both increases **and** is concave downwards. 2
- iii) If the curve passes through  $(1, -2)$  find the equation of the curve. 2

### QUESTION 3

Marks

- a) Find the primitive function of  $\sqrt{x}$  2
- b) Melanie joined a Superannuation Fund, investing \$P at the beginning of every year at 8% p.a. compound interest (compounded yearly).
- i) Write an expression for the amount of her investment at the end of the first year. 1
  - ii) Write an expression for the amount of her investment at the end of the second year. 1
  - iii) Write an expression for the amount of her investment at the end of twenty five years 1
  - iv) If at the end of twenty five years, she wishes to collect \$500,000 calculate the value of \$P to the nearest dollar. 2

c)



The shaded region OAB is bounded by the parabolas  $y = x^2$  and  $y = 2 - x^2$  and the  $x$  axis from  $x = 0$  to  $x = \sqrt{2}$ .

- i) B is the point of intersection of the two parabolas in the first quadrant. Find the co-ordinates of B. 2
- ii) Calculate the area of the shaded region OAB ( 2 dp). 3

**QUESTION 4****Marks**

a) A function is defined by  $y = 3x^2 - 2x^3$

- |      |                                                                        |          |
|------|------------------------------------------------------------------------|----------|
| i)   | Find the co-ordinates of any turning points and determine their nature | <b>3</b> |
| ii)  | Given that there is a point of inflexion, find its co-ordinates.       | <b>1</b> |
| iii) | Sketch the function from $x = -1$ to $x = 2$                           | <b>2</b> |

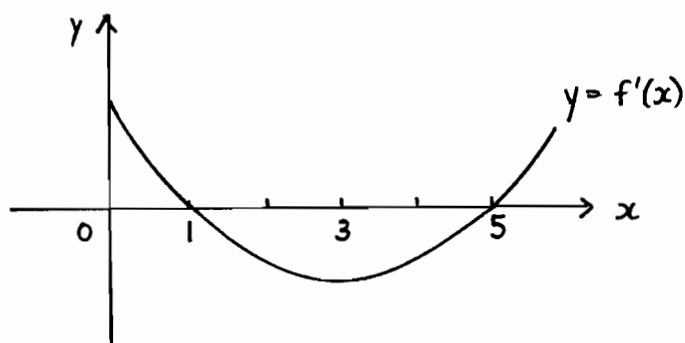
**Note** Your sketch must be neat

Use a ruler to draw the axes

Label all important points

- |     |                                                                                               |          |
|-----|-----------------------------------------------------------------------------------------------|----------|
| iv) | Find the area bounded by the curve $y = 3x^2 - 2x^3$ and the $x$ axis from $x = 0$ to $x = 2$ | <b>4</b> |
|-----|-----------------------------------------------------------------------------------------------|----------|

b) **2**



The diagram shows the graph of the gradient function of the curve  $y = f(x)$ .

For what value of  $x$  does  $f(x)$  have a local minimum?

**QUESTION 5****Marks**

- a)** A cylindrical container closed at both ends is made from a sheet of thin plastic.  
The surface area of the cylinder is  $600\pi$  centimetres<sup>2</sup>.

- i) Show that the height  $h$  of the cylinder is given by the expression: **2**

$$h = \frac{300}{r} - r, \text{ where } r \text{ is the radius.}$$

- ii) Find an expression for the volume  $V$  in terms of  $r$ . **1**

- iii) Find the height of the container if the volume is to be a maximum. **3**

- b)** i) Differentiate  $y = x^3(1+x)^3$  **2**

- ii) Hence, solve  $\frac{dy}{dx} = 0$  **2**

**End of Test**

QUESTION 1

a) i.  $y = x^3 + 4x^2 + 2$   
 $\frac{dy}{dx} = \underline{\underline{3x^2 + 8x}}$

ii.  $y = \frac{3x}{x+2}$

$$u = 3x \quad v = x+2$$

$$u' = 3 \quad v' = 1$$

$$\frac{dy}{dx} = \frac{3(x+2) - 3x(1)}{(x+2)^2}$$

$$= \frac{3x+6-3x}{(x+2)^2}$$

$$= \frac{6}{(x+2)^2}$$

iii.  $y = (2x+1)^4$   
 $\frac{dy}{dx} = 4(2x+1)^3 \times 2$   
 $\frac{dy}{dx} = \underline{\underline{8(2x+1)^3}}$

b)  $y = 4x^3 + x$   
 $\frac{dy}{dx} = 12x^2 + 1$

when  $x=1$ ,  $m_{\text{tangent}} = 12 \times 1^2 + 1$   
 $= \underline{\underline{13}}$

c) i.  $\int x^4 + 3x^2 dx$   
 $= \frac{x^5}{5} + x^3 + c$

ii.  $\int (x-5)(x+4) dx$   
 $= \int x^2 - x - 20 dx$   
 $= \frac{x^3}{3} - \frac{x^2}{2} - 20x + c$

iii.  $\int \frac{x^3 - 3x^4}{x^2} dx$   
 $= \int x - 3x^2 dx$   
 $= \frac{x^2}{2} - x^3 + c$

QUESTION 2

a)  $\int_0^4 f(x) dx = \frac{1}{2} \times \pi \times 1^2$   
 $= \underline{\underline{\frac{\pi}{2}}}$

b)  $\int (2x-1)^5 dx = \frac{(2x-1)^6}{6 \times 2} + c$   
 $= \frac{(2x-1)^6}{12} + c$

c)  $A = 5 \times 2 - \int_0^2 x^2 + 1 dx$   
 $= 10 - \left[ \frac{x^3}{3} + x \right]_0^2$   
 $= 10 - \left[ \frac{2^3}{3} + 2 - 0 \right]$   
 $= 10 - 4\frac{2}{3}$   
 $= \underline{\underline{5\frac{1}{3}}}$   $u^2$

d)  $\frac{dy}{dx} = 3x^2 - 12$

i.  $\frac{d^2y}{dx^2} = \underline{\underline{6x}}$

ii. Increasing:  $\frac{dy}{dx} > 0$   
 $3x^2 - 12 > 0$   
 $3(x+2)(x-2) > 0$   
 $x < -2, x > 2$

Concave down:  $\frac{d^2y}{dx^2} < 0$   
 $6x < 0$   
 $x < 0$

$\therefore$  both increasing and concave down  $\underline{\underline{x < -2}}$

iii.  $y = \int 3x^2 - 12 dx$

$y = x^3 - 12x + c$   
 when  $x=1$ ,  $y=-2$   
 $-2 = 1^3 - 12 \cdot 1 + c$   
 $c = 9$

$\therefore y = x^3 - 12x + 9$

### QUESTION 3

$$\begin{aligned}
 \text{a) } \sqrt{x} &= x^{\frac{1}{2}} \\
 \text{primitive} &= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C \\
 &= \frac{2}{3} x^{\frac{3}{2}} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{b) i. } A_1 &= P(1.08) \\
 \text{ii. } A_2 &= P(1.08)^2 + P(1.08) \\
 &= P(1.08^2 + 1.08) \\
 \text{iii. } A_{25} &= P(1.08^{25} + 1.08^{24} + \dots + 1.08) \\
 \text{iv. } A_{25} &= \frac{P \times 1.08(1.08^{25} - 1)}{1.08 - 1} \\
 500\,000 &= \frac{P \times 1.08(1.08^{25} - 1)}{0.08} \\
 P &= \underline{\underline{\$6333}} \text{ (nearest dollar)}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) i. } x^2 &= 2 - x^2 \\
 2x^2 &= 2 \\
 x^2 &= 1 \\
 x &= 1 \text{ (1st quadrant)} \\
 y &= 1^2 \\
 \therefore B(1, 1)
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } A &= \int_0^1 x^2 dx + \int_1^{\sqrt{2}} 2 - x^2 dx \\
 &= \left[ \frac{x^3}{3} \right]_0^1 + \left[ 2x - \frac{x^3}{3} \right]_1^{\sqrt{2}} \\
 &= \left[ \frac{1^3}{3} - 0 \right] + \left[ (2\sqrt{2} - \frac{\sqrt{2}^3}{3}) - (2 \times 1 - \frac{1^3}{3}) \right] \\
 &= \frac{1}{3} + \left[ 2\sqrt{2} - \frac{2\sqrt{2}}{3} - (2 - \frac{1}{3}) \right]
 \end{aligned}$$

$$\therefore A = \underline{\underline{0.55}} \text{ (2 dp)}$$

### QUESTION 4

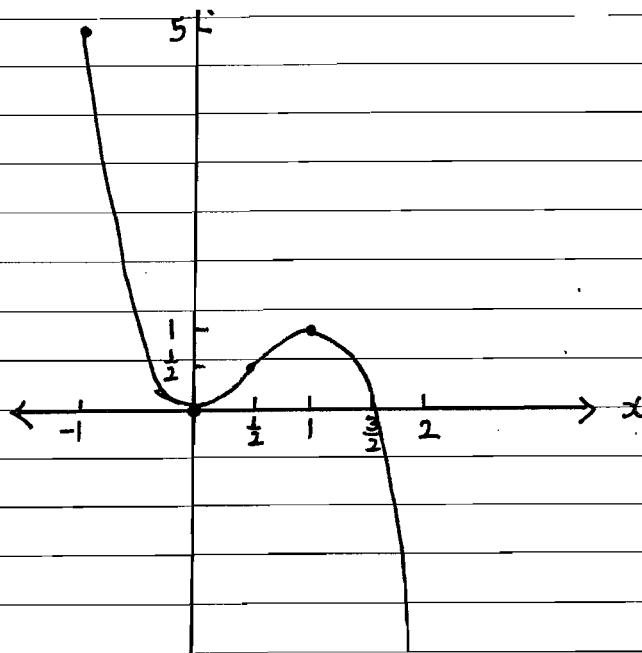
$$\begin{aligned}
 \text{a) } y &= 3x^2 - 2x^3 \\
 \frac{dy}{dx} &= 6x - 6x^2 \\
 \frac{d^2y}{dx^2} &= 6 - 12x
 \end{aligned}$$

$$\begin{aligned}
 \text{i. Stat pts: } \frac{dy}{dx} &= 0 \\
 6x - 6x^2 &= 0 \\
 6x(1-x) &= 0 \\
 x &= 0, 1 \\
 \text{When } x=0, y &= 0 \\
 \frac{d^2y}{dx^2} &> 0 \\
 \therefore \underline{\underline{\text{min at } (0, 0)}}
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x=1, y &= 1 \\
 \frac{d^2y}{dx^2} &< 0
 \end{aligned}$$

$$\therefore \underline{\underline{\text{max at } (1, 1)}}$$

$$\begin{aligned}
 \text{ii. Inflexion: } \frac{d^2y}{dx^2} &= 0 \\
 6 - 12x &= 0 \\
 12x &= 6 \\
 x &= \frac{1}{2} \\
 \therefore \underline{\underline{\text{inflexion at } (\frac{1}{2}, \frac{1}{2})}}
 \end{aligned}$$



$$\begin{aligned}
 \text{iv. } A &= \int_0^{\frac{3}{2}} 3x^2 - 2x^3 dx + \left| \int_{\frac{3}{2}}^2 3x^2 - 2x^3 dx \right| \\
 &= \left[ x^3 - \frac{x^4}{2} \right]_0^{\frac{3}{2}} + \left| \left[ x^3 - \frac{x^4}{2} \right]_{\frac{3}{2}}^2 \right| \\
 &= \left[ \left( \frac{3}{2} \right)^3 - \frac{1}{2} \times \left( \frac{3}{2} \right)^4 - 0 \right] + \left| \left[ 2^3 - \frac{2^4}{2} \right] - \left[ \left( \frac{3}{2} \right)^3 - \frac{1}{2} \times \left( \frac{3}{2} \right)^4 \right] \right| \\
 &= \frac{27}{32} + \left| -\frac{27}{32} \right|
 \end{aligned}$$

$$\therefore A = \underline{\underline{1 \frac{11}{16}}} u^2 \quad (1.6875)$$

b)  $f'(x) = 0$  at  $x = 1$  and  $5$   
i.e. stationary points on  $f(x)$

when  $x < 1$ ,  $f'(x) > 0$

i.e. increasing

when  $1 < x < 5$ ,  $f'(x) < 0$

i.e. decreasing

when  $x > 5$ ,  $f'(x) > 0$

i.e. increasing

$\therefore$  min when  $x = 5$

### QUESTION 5

$$\begin{aligned}
 \text{a) i. } SA &= 2\pi r^2 + 2\pi rh \\
 600\pi &= 2\pi r^2 + 2\pi rh \\
 300 &= r^2 + rh \\
 rh &= 300 - r^2 \\
 h &= \frac{300 - r^2}{r}
 \end{aligned}$$

$$\therefore h = \frac{300}{r} - r$$

$$\begin{aligned}
 \text{ii. } V &= \pi r^2 h \\
 &= \pi r^2 (300 - r)
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. } \frac{dV}{dr} &= 300\pi - 3\pi r^2 \\
 \frac{d^2V}{dr^2} &= -6\pi r
 \end{aligned}$$

$$\text{max } V : \frac{dV}{dr} = 0$$

$$300\pi - 3\pi r^2 = 0$$

$$3\pi (100 - r^2) = 0$$

$$r = 10 \text{ only } (r > 0)$$

$$\text{when } r = 10, \frac{d^2V}{dr^2} < 0$$

$$\therefore \text{max } V \text{ when } r = 10$$

$$\therefore \text{height} = \frac{300}{10} - 10$$

$$\therefore h = \underline{\underline{20 \text{ cm}}}$$

$$\text{b) i. } y = x^3(1+x)^3$$

$$u = x^3 \quad v = (1+x)^3$$

$$u' = 3x^2 \quad v' = 3(1+x)^2$$

$$\begin{aligned}
 \frac{dy}{dx} &= 3x^2(1+x)^3 + 3x^3(1+x)^2 \\
 &= 3x^2(1+x)^2[(1+x) + x] \\
 &= 3x^2(1+x)^2(1+2x)
 \end{aligned}$$

$$\text{ii. } 0 = 3x^2(1+x)^2(1+2x)$$

$$\therefore x = \underline{\underline{0, -1, -\frac{1}{2}}}$$